

Preconditioning without a preconditioner: faster ridge-regression and Gaussian sampling with randomized block Krylov subspace methods

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Motivation

- › Many data-science tasks require solving large, regularized linear systems.
- › Examples include ridge regression and sampling high-dimensional Gaussians.
- › Idea:
 - Develop a randomized block conjugate gradient (block-CG) method that improves convergence—achieving the benefits of preconditioning without explicitly constructing one.

Problem Formulation

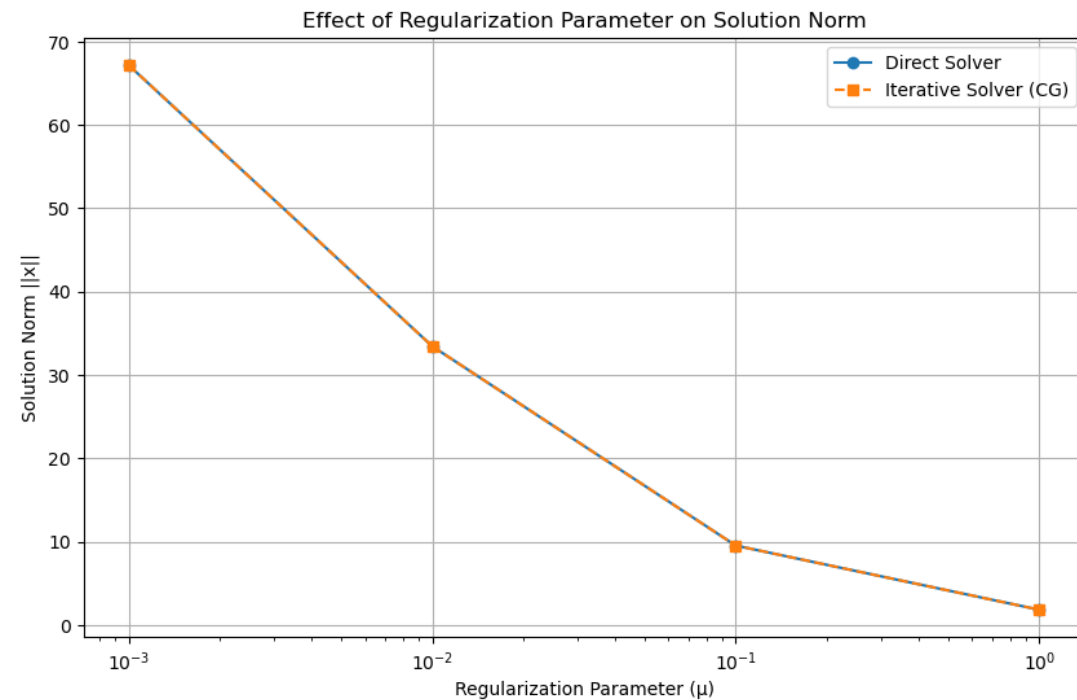
- › Equation (1): Solving the regularized linear system

$$\mathbf{A}_\mu \mathbf{x} = \mathbf{b}, \quad \mathbf{A}_\mu := \mathbf{A} + \mu \mathbf{I}, \quad (1)$$

where $\mathbf{A} \in \mathbb{R}^{d \times d}$ is symmetric positive definite and $\mu \geq 0$ is a critical task across the computational sciences. Systems of the form (1) arise in a variety of settings, but we are particularly motivated by two tasks in machine learning and data science: solving ridge-regression problems and sampling Gaussian vectors.

- › Explanation:
 - This formulation covers both ridge regression and the systems encountered when sampling Gaussian vectors.

Regularization Pattern Parameter



```
x, info = spla.cg(regularized_matrix, b, tol=tol, maxiter=max_iter)
```

Comparison of Direct and Iterative Solvers for Regularized Systems $(A + \mu I)x = b$

μ	Direct Norm	Iterative Norm	Difference	Direct Residual	Iterative Residual
1.000e-03	6.717279e+01	6.717279e+01	2.226790e-10	2.395620e-13	2.239940e-09
1.000e-02	3.344389e+01	3.344389e+01	3.658206e-10	1.094057e-13	3.282182e-09
1.000e-01	9.528185e+00	9.528185e+00	6.212907e-10	3.246131e-14	9.234232e-09
1.000e+00	1.827953e+00	1.827953e+00	1.519871e-08	6.589854e-15	8.200025e-09

Ridge Regression Formulation

› Objective & Normal Equations:

Ridge regression. Given a data matrix $\mathbf{Z} \in \mathbb{R}^{n \times d}$, a source term $\mathbf{f} \in \mathbb{R}^n$, and a regularization parameter $\mu \geq 0$, the ridge regression problem¹ is

$$\min_{\mathbf{x} \in \mathbb{R}^d} \|\mathbf{Z}\mathbf{x} - \mathbf{f}\|^2 + \mu \|\mathbf{x}\|^2. \quad (2)$$

A direct computation shows that the solution to (2) is also the solution to

$$(\mathbf{Z}^T \mathbf{Z} + \mu \mathbf{I})\mathbf{x} = \mathbf{Z}^T \mathbf{f}; \quad (3)$$

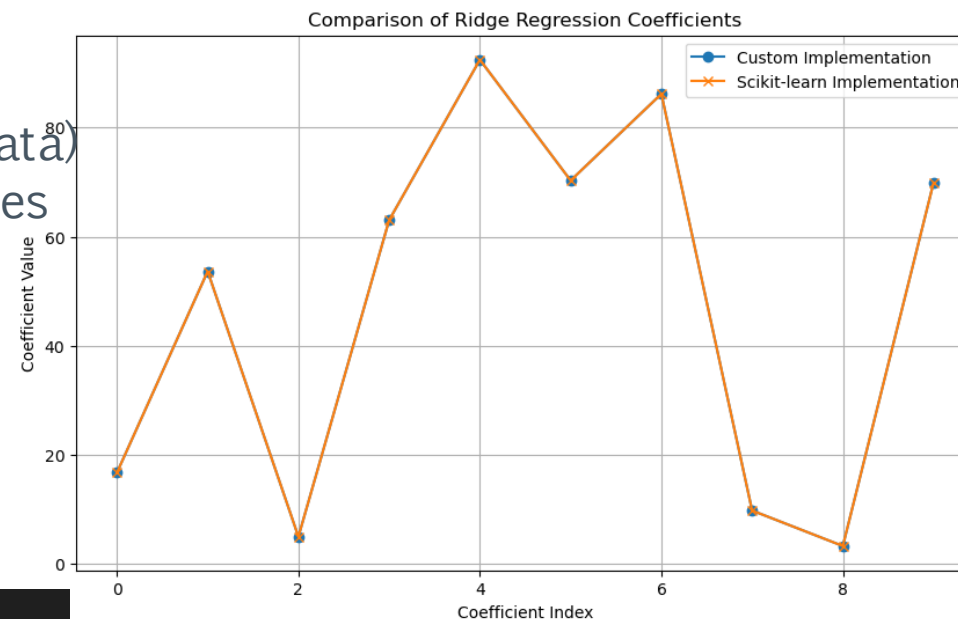
i.e. a system of the form (1) with $\mathbf{A} = \mathbf{Z}^T \mathbf{Z}$ and $\mathbf{b} = \mathbf{Z}^T \mathbf{f}$. In a number of applications we are interested in the whole regularization path; i.e. $\mathbf{x} = \mathbf{x}_\mu$ for all values $\mu \geq 0$; e.g. in order to perform cross-validation to select a value of μ to use for future predictions.

In ridge regression, we penalize large coefficients (controlled by μ) to avoid overfitting. When $\mu=0$, this is the standard least-squares problem.

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Ridge Regression Coefficients

The first term minimizes prediction error (fit to data)
The second term penalizes large coefficient values (regularization)



Custom Ridge Regression coefficients:

```
[16.76598699 53.65898412  5.02093527 63.04853154 92.44608299 70.3107064
 86.18115355  9.81635392  3.3184875  69.93150662]
```

Scikit-learn Ridge Regression coefficients:

```
[16.76598699 53.65898412  5.02093527 63.04853154 92.44608299 70.3107064
 86.18115355  9.81635392  3.3184875  69.93150662]
```

Difference between custom and scikit-learn coefficients:

```
5.344037511214127e-14
```

Gaussian Sampling

› Sampling Gaussian

Sampling Gaussians. If $\mathbf{b}$ has independent standard normal entries, then $\boldsymbol{\mu} + \mathbf{A}^{1/2}\mathbf{b}$ is a Gaussian vector with mean $\boldsymbol{\mu}$ and covariance $\mathbf{A}$. In order to approximate $\mathbf{A}^{1/2}\mathbf{b}$, it is common to make use of the identity

$$\mathbf{A}^{1/2}\mathbf{b} = \frac{2}{\pi} \int_0^\infty \mathbf{A}(\mathbf{A} + z^2\mathbf{I})^{-1}\mathbf{b} \, dz. \quad (4)$$

The term $(\mathbf{A} + z^2\mathbf{I})^{-1}\mathbf{b}$ in the integrand of (4) is of the form (1) with $\mu = z^2$. Often we are interested in sampling several Gaussian vectors with the same covariance matrix.

To sample from a gaussian, $N(\boldsymbol{\mu}, \mathbf{A})$, if $\mathbf{b} \sim N(0, \mathbf{I})$ then $N(0, \mathbf{I}) \sim N(0, \mathbf{I})$ then

$$\boldsymbol{\mu} + \mathbf{A}^{1/2}\mathbf{b} \sim N(\boldsymbol{\mu}, \mathbf{A}).$$

The equation expresses the matrix square root via an integral representation.

Gaussian

- › these visual patterns directly reflect the covariance structure of the data.

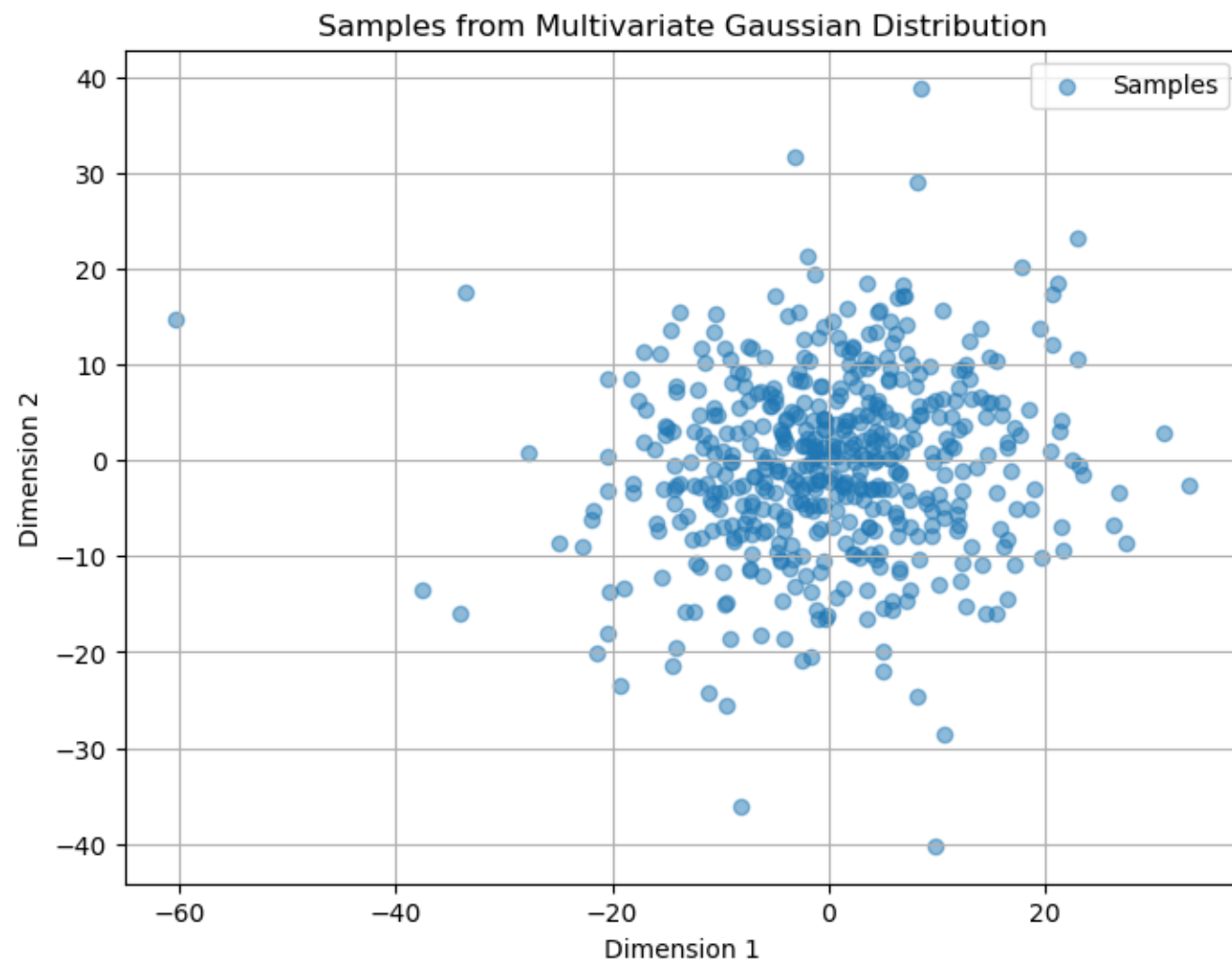
- › $p(x) = (1/((2\pi)^{n/2} |\Sigma|^{1/2})) * \exp(-1/2 (x-\mu)^T \Sigma^{-1} (x-\mu))$

- › Σ 's properties - its eigenvectors determine the principal axes directions, while its eigenvalues determine their lengths.

- › $x = \mu + A^{1/2}z$

- › block-Lanczos approximation

- › $A^{1/2}b \approx \sum \beta_i q_i q_i^T b$



Computational Assumptions & Notation

- › A is accessed only via matrix-vector or block matrix-vector products.

Eigenvalues: $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d$

Condition Number: $\kappa(M) = \lambda_1(M)/\lambda_d(M)$

Spectral Norm: $\|M\| = \lambda_1(M)$

M -norm of x : $\|x\|_M = \|M^{1/2}x\|$

Krylov Subspace Methods

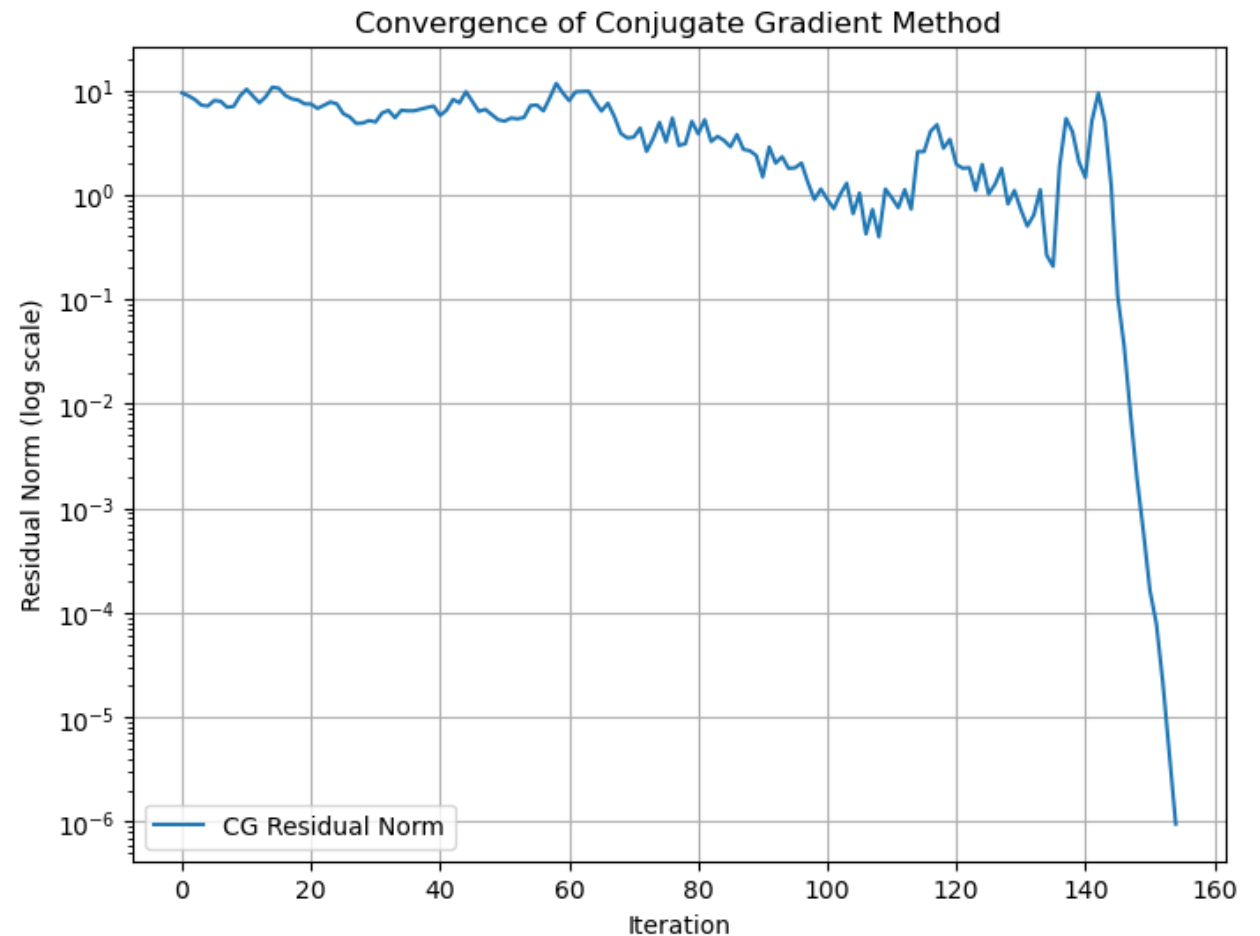
- › These subspaces capture most of the action of A and are used to build iterative solvers.

Krylov Subspace Methods (KSM) make use of the Krylov subspace²

$$\mathcal{K}_t(\mathbf{A}, \mathbf{b}) := \text{span}\{\mathbf{b}, \mathbf{A}\mathbf{b}, \dots, \mathbf{A}^{t-1}\mathbf{b}\}, \quad (5)$$

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Convergence CG



Norm of the difference between CG and direct solution: $1.1894033922928449e-08$

PRECONDITIONED CONJUGATE GRADIENT (PCG)

PCG Iterate definition:

Definition 2.1. The t -th PCG iterate corresponding to a positive definite preconditioner $\mathbf{P}_\mu$ is defined as

$$\text{pcg}_t(\mu) := \underset{\mathbf{x} \in \mathcal{K}_t(\mathbf{A}_\mu, \mathbf{b}; \mathbf{P}_\mu)}{\text{argmin}} \quad \|\mathbf{A}_\mu^{-1} \mathbf{b} - \mathbf{x}\|_{\mathbf{A}_\mu}.$$

The iterate $\text{pcg}_t(\mu)$ can be computed using $t - 1$ matrix-vector products with $\mathbf{A}$ (and t products with $\mathbf{P}_\mu^{-1}$). We discuss implementation details in [Appendix A](#).

The simplest choice of preconditioner is $\mathbf{P}_\mu = \mathbf{I}$, which yields the standard conjugate gradient algorithm (CG) ([Hestenes & Stiefel, 1952](#)).

The preconditioner $\mathbf{P}_\mu$ is used to improve the convergence rate of CG.

Corollary 2.3. Let $\mathbf{P}_\mu$ be any preconditioner and let

$$\varepsilon_t(\mu) := 2 \exp \left(- \frac{2t}{\sqrt{\kappa(\mathbf{P}_\mu^{-1/2} \mathbf{A}_\mu \mathbf{P}_\mu^{-1/2})}} \right).$$

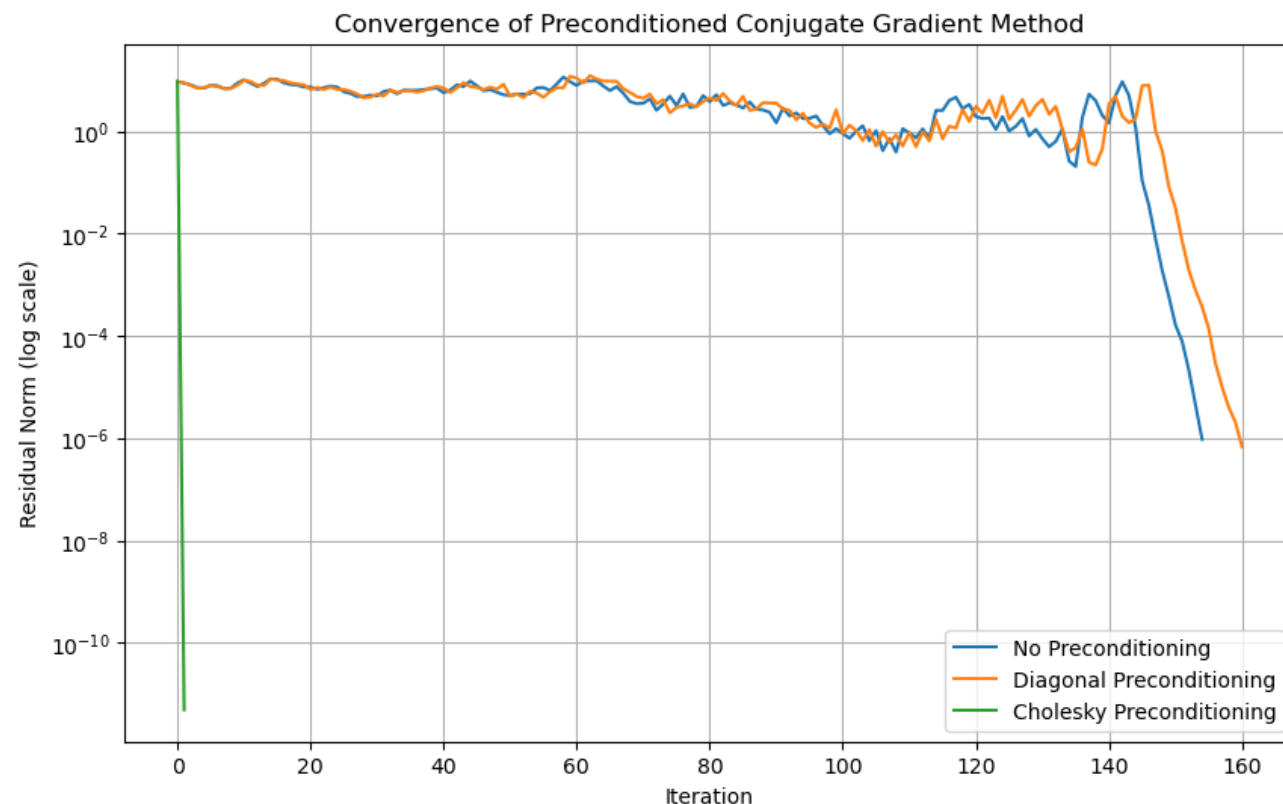
Then the t -th *preconditioned-CG* iterate corresponding to the preconditioner $\mathbf{P}_\mu$ satisfies

$$\frac{\|\mathbf{A}_\mu^{-1} \mathbf{b} - \text{pcg}_t(\mu)\|_{\mathbf{A}_\mu}}{\|\mathbf{A}_\mu^{-1} \mathbf{b}\|_{\mathbf{A}_\mu}} \leq \varepsilon_t(\mu).$$

Corollary 2.3 implies that if we can find $\mathbf{P}_\mu$ so that $\kappa(\mathbf{P}_\mu^{-1/2} \mathbf{A}_\mu \mathbf{P}_\mu^{-1/2})$ is small, then PCG converges rapidly. The choice of $\mathbf{P}_\mu$ minimizing this condition number is of course $\mathbf{P}_\mu = \mathbf{A}_\mu^{-1}$, but this is not a practical preconditioner; if we knew $\mathbf{A}_\mu^{-1}$ we could easily compute the solution $\mathbf{A}_\mu^{-1} \mathbf{b}$. Thus, finding a suitable choice of $\mathbf{P}_\mu$, which balances improvements to the convergence of PCG with the cost of building/applying $\mathbf{P}_\mu$ is critical.

Convergence CG vs PCG

- Accelerated Convergence: The PCG line drops more steeply than standard CG, demonstrating the effectiveness of preconditioning.
- Condition Number Improvement: The slope of the line directly reflects how well the preconditioner has improved the spectral properties of the system.
- Eigenvalue Clustering: Any plateaus in the convergence curve may indicate clusters of eigenvalues that influence the convergence behavior.
- Preconditioning Quality: The gap between PCG and standard CG lines quantifies the effectiveness of the chosen preconditioner.
- Iteration Efficiency: PCG typically achieves the same accuracy with significantly fewer iterations than standard CG.



Deflated Preconditioner

› Deflated Preconditioner

In particular, one can form the preconditioner

$$\mathbf{P}_\mu := \frac{1}{\theta + \mu} \mathbf{U}(\mathbf{D} + \mu \mathbf{I}) \mathbf{U}^\top + (\mathbf{I} - \mathbf{U} \mathbf{U}^\top), \quad (8)$$

and it is not hard to verify that

$$\mathbf{P}_\mu^{-1} = (\theta + \mu) \mathbf{U}(\mathbf{D} + \mu \mathbf{I})^{-1} \mathbf{U}^\top + (\mathbf{I} - \mathbf{U} \mathbf{U}^\top). \quad (9)$$

Here $\theta > 0$ is a parameter that must be chosen along with the factorization $\mathbf{U} \mathbf{D} \mathbf{U}^\top$.

Applying $\mathbf{P}_\mu^{-1}$ to a vector requires $O(dr)$ arithmetic operations, which is relatively cheap if $r \ll d$. Hence, so long as a reasonable factorization $\mathbf{U} \mathbf{D} \mathbf{U}^\top$ can be obtained, (8) can be used as a preconditioner; see (Frank & Vuik, 2001; Gutknecht, 2012) and the references within.

This preconditioner “deflates” the large eigenvalues, improving the condition number of $\mathbf{A}_\mu$

Nystrom Preconditioning

› Equation 11 & 13:

In particular, it's reasonable to take $\mathbf{U}\mathbf{D}\mathbf{U}^\top$ as the eigendecomposition of the Nyström approximation

$$\mathbf{A}\langle\mathbf{K}_s\rangle := (\mathbf{A}\mathbf{K}_s)(\mathbf{K}_s^\top\mathbf{A}\mathbf{K}_s)^\dagger(\mathbf{K}_s^\top\mathbf{A}), \quad (11)$$

where $\mathbf{\Omega} \in \mathbb{R}^{d \times \ell}$ is a matrix of independent standard normal random variables and

$$\mathbf{K}_s := [\mathbf{\Omega} \mathbf{A} \mathbf{\Omega} \cdots \mathbf{A}^{s-1} \mathbf{\Omega}] \in \mathbb{R}^{d \times (s\ell)}. \quad (12)$$

This variant of the Nyström approximation is among the most powerful randomized low-rank approximation algorithms, and can be implemented using s matrix-loads (Tropp & Webber, 2023).

This randomized low-rank approximation helps to form an implicit preconditioner that improves convergence without explicitly computing one.

Most related to the content of this paper is the theoretical analysis of (Frangella et al., 2023), which proves that if $s = 1$, $\theta = \lambda_\ell(\mathbf{A}\langle\mathbf{\Omega}\rangle)$, and the sketching dimension ℓ is on the order of the effective dimension

$$d_{\text{eff}}(\mu) := \text{tr}(\mathbf{A}\mathbf{A}_\mu^{-1}) = \sum_{i=1}^d \frac{\lambda_i}{\lambda_i + \mu}, \quad (13)$$

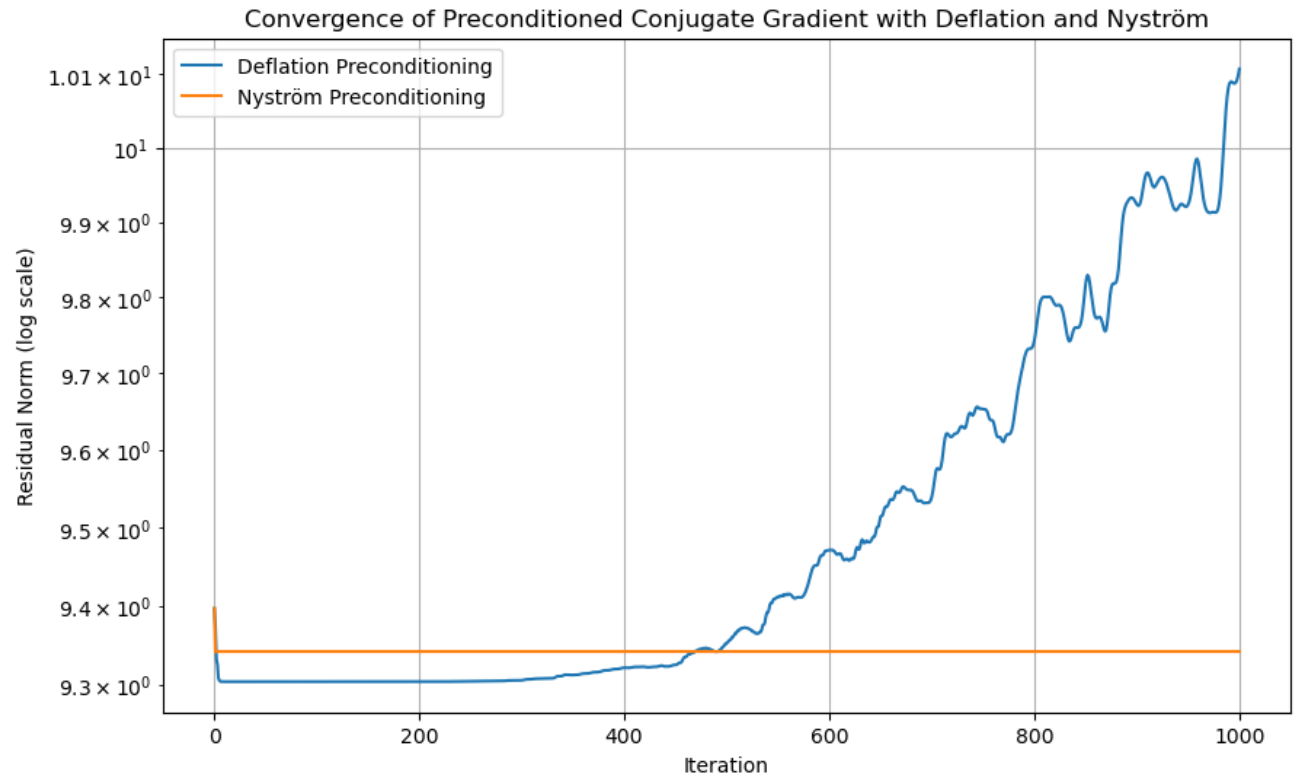
then Nyström PCG converges in at most $t = O(\log(1/\varepsilon))$ iterations; i.e. independent of any spectral properties of $\mathbf{A}$. Our analysis makes use of the same general techniques as (Frangella et al., 2023), but is applicable when $s > 1$ as well as if $\mu = 0$. We discuss the theoretical bounds of (Frangella

PCG WITH DEFLATION AND NYSTRÖM

For deflation, we compute the eigen-decomposition of A and build a preconditioner that ‘deflates’ the largest eigenvalues.

For Nyström preconditioning, we use a randomized sampling of columns to form an approximation of A .

We then solve the system using PCG with these preconditioners and plot the convergence. The comparison shows how these techniques reduce the number of iterations required.”



Deflation and Nyström preconditioning enhance standard PCG by specifically addressing the problematic eigenvalues of A . The Nyström method offers efficient low-rank approximation.

The deflation method explicitly removes the effect of the large eigenvalues, while the Nyström method uses randomness to efficiently approximate the matrix. The convergence plots illustrate that both methods improve the solver's performance.

AUGMENTED BLOCK CG METHOD

Block Krylov Subspace

In this paper, we advocate for the use of block-KSMs for solving (1). Given a matrix $\mathbf{B} \in \mathbb{R}^{d \times m}$ (typically $m \ll d$) with columns $\mathbf{b}^{(1)}, \dots, \mathbf{b}^{(m)}$, the block Krylov subspace is

$$\mathcal{K}_t(\mathbf{A}, \mathbf{B}) := \mathcal{K}_t(\mathbf{A}, \mathbf{b}^{(1)}) + \dots + \mathcal{K}_t(\mathbf{A}, \mathbf{b}^{(m)}). \quad (14)$$

That is, $\mathcal{K}_t(\mathbf{A}, \mathbf{B})$ is the space consisting of all linear combinations of vectors in $\mathcal{K}_t(\mathbf{A}, \mathbf{b}^{(1)}), \dots, \mathcal{K}_t(\mathbf{A}, \mathbf{b}^{(m)})$.

This naturally gives rise to the block-CG algorithm (O’Leary, 1980).

› Block-CG Iterate

Definition 3.1. Let $\mathbf{B} = [\mathbf{b}^{(1)} \dots \mathbf{b}^{(m)}]$. The t -th block-CG iterates are defined as

$$\text{bcg}_t^{(i)}(\mu) := \underset{\mathbf{x} \in \mathcal{K}_t(\mathbf{A}, \mathbf{B})}{\operatorname{argmin}} \|\mathbf{A}_\mu^{-1} \mathbf{b}^{(i)} - \mathbf{x}\|_{\mathbf{A}_\mu}. \quad (15)$$

The block-CG iterates $\text{bcg}_t^{(1)}(\mu), \dots, \text{bcg}_t^{(m)}(\mu)$ can be simultaneously computed using $t - 1$ block matrix-vector products with $\mathbf{A}$. We discuss implementation and costs further in Section 3.2.

By augmenting the starting block with extra vectors (e.g., from a random matrix Ω , the method effectively “preconditions” the system implicitly.

MAIN THEORETICAL RESULT –

Key Error Guarantee:

Theorem 3.2. Fix any matrix $\Omega \in \mathbb{R}^{d \times m}$ and let $\mathbf{P} = (\mathbf{I} + \mathbf{X})^{-1}$ be any preconditioner where $\text{range}(\mathbf{X}) \subseteq \mathcal{K}_{s+1}(\mathbf{A}, \Omega)$. Define the augmented starting block $\mathbf{B} = [\mathbf{b} \ \Omega]$. Then, for any $t \geq s$, the t -th *block-CG* iterate is related to the $(t - s)$ -th *preconditioned-CG* iterate corresponding to the preconditioner $\mathbf{P}_\mu$ in that

$$\|\mathbf{A}_\mu^{-1}\mathbf{b} - \text{bcg}_t^{(1)}(\mu)\|_{\mathbf{A}_\mu} \leq \|\mathbf{A}_\mu^{-1}\mathbf{b} - \text{pcg}_{t-s}(\mu)\|_{\mathbf{A}_\mu}.$$

In particular, when $\mathbf{U}\mathbf{D}\mathbf{U}^\top = \mathbf{A}\langle\mathbf{K}_s\rangle$, then the deflation preconditioner $\mathbf{P}_\mu$ defined in (8) has the form

$$\mathbf{P}^{-1} = \mathbf{I} + \mathbf{X}, \text{ where } \text{range}(\mathbf{X}) \subseteq \mathcal{K}_s(\mathbf{A}, \Omega). \quad (16)$$

Augmentation Benefit

Remark 3.3. Theorem 3.2 asserts that augmented block-CG automatically performs no worse than Nyström PCG (with the best choice of s and θ) after the same number of matrix-loads.⁴ This explains the performance observed in Figure 1.

This theorem shows that our augmented block-CG method is at least as good as (and often better than) the corresponding PCG method with an explicit Nystrom preconditioner.

CONVERGENCE BOUNDS FOR NYSTRÖM PRECONDITIONING

Error Bound for Nystrom Approximation

We begin by recalling a error guarantee for Nyström low-rank approximation, which compares the Nyström error $\|\mathbf{A} - \mathbf{A}\langle \mathbf{K}_s \rangle\|$ to the error $\|\mathbf{A} - \llbracket \mathbf{A} \rrbracket_r\| = \lambda_{r+1}$ of the best possible rank- r approximation to $\mathbf{A}$.

Theorem 4.2 (Theorem 9.1 in (Tropp & Webber, 2023)). Suppose $\Omega \in \mathbb{R}^{d \times (r+p)}$ is a random Gaussian matrix and define

$$\mathbf{K}_s := [\Omega \mathbf{A} \Omega \cdots \mathbf{A}^{s-1} \Omega]. \quad (20)$$

Then, if $p \geq 2$,

$$\begin{aligned} & \log \left(\frac{\mathbb{E} \|\mathbf{A} - \mathbf{A}\langle \mathbf{K}_s \rangle\|^2}{\lambda_{r+1}^2} \right) \\ & \leq \frac{1}{8(s - \frac{3}{2})^2} \log \left(4 + \frac{4r}{p-1} \sum_{i>r} \frac{\lambda_i^2}{\lambda_{r+1}^2} \right)^2. \end{aligned}$$

› Condition Number bound:

Theorem 4.3. Let $\mathbf{P}_\mu$ be the Nyström preconditioner (8) corresponding to the Nyström approximation $\mathbf{A}\langle \mathbf{K} \rangle$ for any $\mathbf{K}$ and shift parameter $\theta \geq 0$. Then

$$\begin{aligned} & \kappa(\mathbf{P}_\mu^{-1/2} \mathbf{A} \mathbf{P}_\mu^{-1/2}) \\ & \leq (\theta + \mu + \|\mathbf{A} - \mathbf{A}\langle \mathbf{K} \rangle\|) \left(\frac{1}{\theta + \mu} + \frac{1}{\lambda_d + \mu} \right). \end{aligned}$$

To prove our main convergence guarantee for our augmented block-CG, we derive a new bound for Nyström preconditioning.

Under suitable conditions (e.g. $p \geq 2$, with appropriate s), with high probability:

Theorem 4.4. Let $\Omega \in \mathbb{R}^{d \times (r+p)}$ be a random Gaussian matrix and define

$$\mathbf{K}_s := [\Omega \mathbf{A} \Omega \cdots \mathbf{A}^{s-1} \Omega]. \quad (21)$$

So these results guarantee that with a properly chosen randomized preconditioner, the condition number is bounded, ensuring rapid convergence.

4.1. Implicit preconditioning with block KSMs

We begin with a key observation.

Theorem 4.1. Suppose $\mathbf{P}_\mu = (\mathbf{I} + \mathbf{X})^{-1}$ for some $\mathbf{X}$ with $\text{range}(\mathbf{X}) \subseteq \mathcal{K}_{s+1}(\mathbf{A}_\mu, \Omega)$. Then,

$$\mathcal{K}_t(\mathbf{A}_\mu, \mathbf{b}; \mathbf{P}_\mu) \subseteq \mathcal{K}_t(\mathbf{A}, \mathbf{b}) + \mathcal{K}_{t+s}(\mathbf{A}, \Omega).$$

MAIN CONVERGENCE GUARANTEE FOR AUGMENTED BLOCK- CG

As follows:

Augmented Block-CG Convergence

We can now state our main convergence guarantee for augmented block-CG. This result is a simple consequence of our condition number bound [Theorem 4.4](#) and the error guarantee [Corollary 2.3](#) for PCG.

Theorem 4.5. *Let $\Omega \in \mathbb{R}^{d \times (r+p)q}$, where $q \geq \log(1/\delta)/\log(100)$, be a random Gaussian matrix. Suppose that $p \geq 2$,*

$$s \geq \frac{5}{2} + \frac{1}{4} \log \left(4 + \frac{4r}{p-1} \sum_{i>r} \frac{\lambda_i^2}{\lambda_{r+1}^2} \right),$$

and

$$\varepsilon_t(\mu) := 2 \exp \left(- \frac{t-s}{\sqrt{(\lambda_{r+1} + \mu)/(\lambda_d + \mu)}} \right).$$

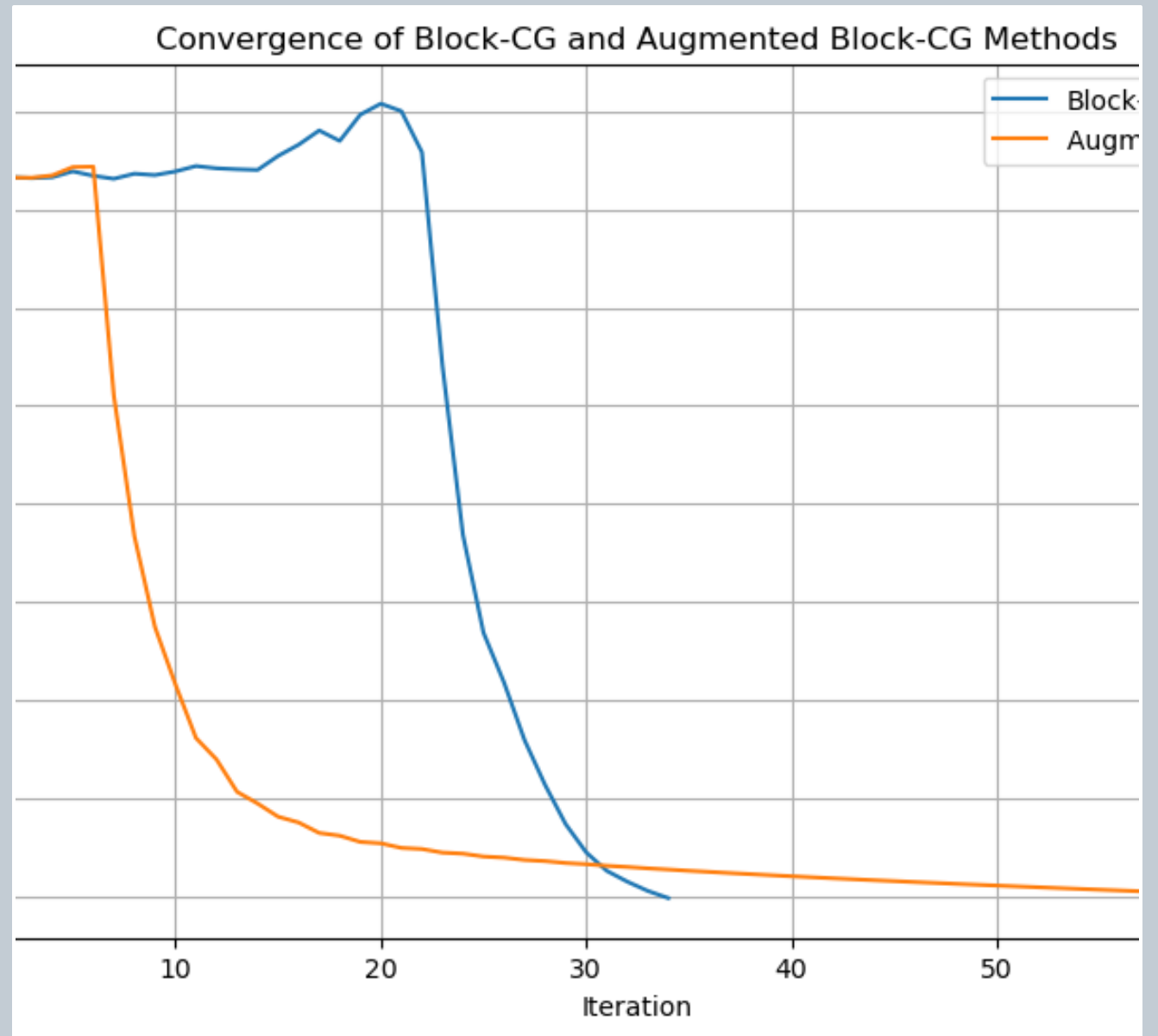
Then, the block-CG iterate $\text{bcg}_t^{(1)}(\mu)$ satisfies, with probability at least $1 - \delta$,

$$\left\{ \forall \mu : \frac{\|\mathbf{A}_\mu^{-1} \mathbf{b} - \text{bcg}_t^{(1)}(\mu)\|_{\mathbf{A}_\mu}}{\|\mathbf{A}_\mu^{-1} \mathbf{b}\|_{\mathbf{A}_\mu}} \leq \varepsilon_t(\mu) \right\}.$$

After a small “burn-in” period (approximately $O(\log(d))$ iterations), the error decays exponentially at a rate determined by the eigenvalue gap.

CONVERGENCE OF BLOCK CG & AUGMENTED CG

- Augmented method consistently achieves lower residuals in fewer iterations,
- It indicates that the augmentation (adding extra random vectors)
 - enhances the quality of the Krylov subspace
 - speeds up convergence.
 - Similar accuracy eventually
 - Exp Error reduction



COMPUTATIONAL COSTS – BLOCK- LANCZOS IMPLEMENTATION

Block-Lanczos Recurrence & Block-CG Iterate Expression

The block-Lanczos algorithm (which we describe explicitly in [Appendix A](#)) applied to $(\mathbf{A}, \mathbf{B})$ for t iterations produces $\mathbf{Q}_t$ with orthonormal columns and $\mathbf{T}_t$ with bandwidth $2m + 1$ satisfying

$$\text{range}(\mathbf{Q}_t) = \mathcal{K}_t(\mathbf{A}, \mathbf{B}), \quad \mathbf{Q}_t^\top \mathbf{A} \mathbf{Q}_t = \mathbf{T}_t. \quad (17)$$

A standard computation (see [Appendix B](#)) reveals that

$$\text{bcg}_k^{(1)}(\mu) = \|\mathbf{b}\| \mathbf{Q}_t (\mathbf{T}_t + \mu \mathbf{I})^{-1} \mathbf{e}_1, \quad (18)$$

where $\mathbf{e}_1 = [1, 0, \dots, 0]^\top$. In particular, since $\mathcal{K}_t(\mathbf{A}_\mu, \mathbf{B}) = \mathcal{K}_t(\mathbf{A}, \mathbf{B})$ for any scalar μ , (18) can be used to compute the block-CG iterate for *multiple values of μ* using the same $\mathbf{Q}_t$ and $\mathbf{T}_t$.

› Cost

Theorem 3.7. Suppose $\mathbf{B} \in \mathbb{R}^{d \times m}$. The block-Lanczos algorithm applied to $(\mathbf{A}, \mathbf{B})$ for t iterations produces $\mathbf{Q}_t$ and $\mathbf{T}_t$ using:

- $t - 1$ matrix-loads of $\mathbf{A}$ (for a total of $m(t - 1)$ total matrix-products)
- $O(dm^2t^2)$ floating point operations operations, and
- $O(dmt)$ storage.

Subsequently, for any $\mu \geq 0$, the t -th block-CG iterates $\text{bcg}_t^{(1)}(\mu)$, can be computed using an additional $O(dmt + m^3t)$ floating point operations operations.

We emphasize that after the block Lanczos algorithm has been run, the **cost** to compute the block-CG iterates for multiple values of μ is relatively small. This is in contrast to Nyström PCG, which is not particularly well suited for solving (1) for multiple values of μ ; while the Nyström approximation can be reused, PCG must be re-run and new products with $\mathbf{A}$ computed.

The approach efficiently reuses the computed Krylov basis to handle multiple regularization parameters with little extra cost.

CONVERGENCE ANALYSIS

The theoretical error bounds for Block-CG show exponential convergence with rate determined by the square root of the condition number

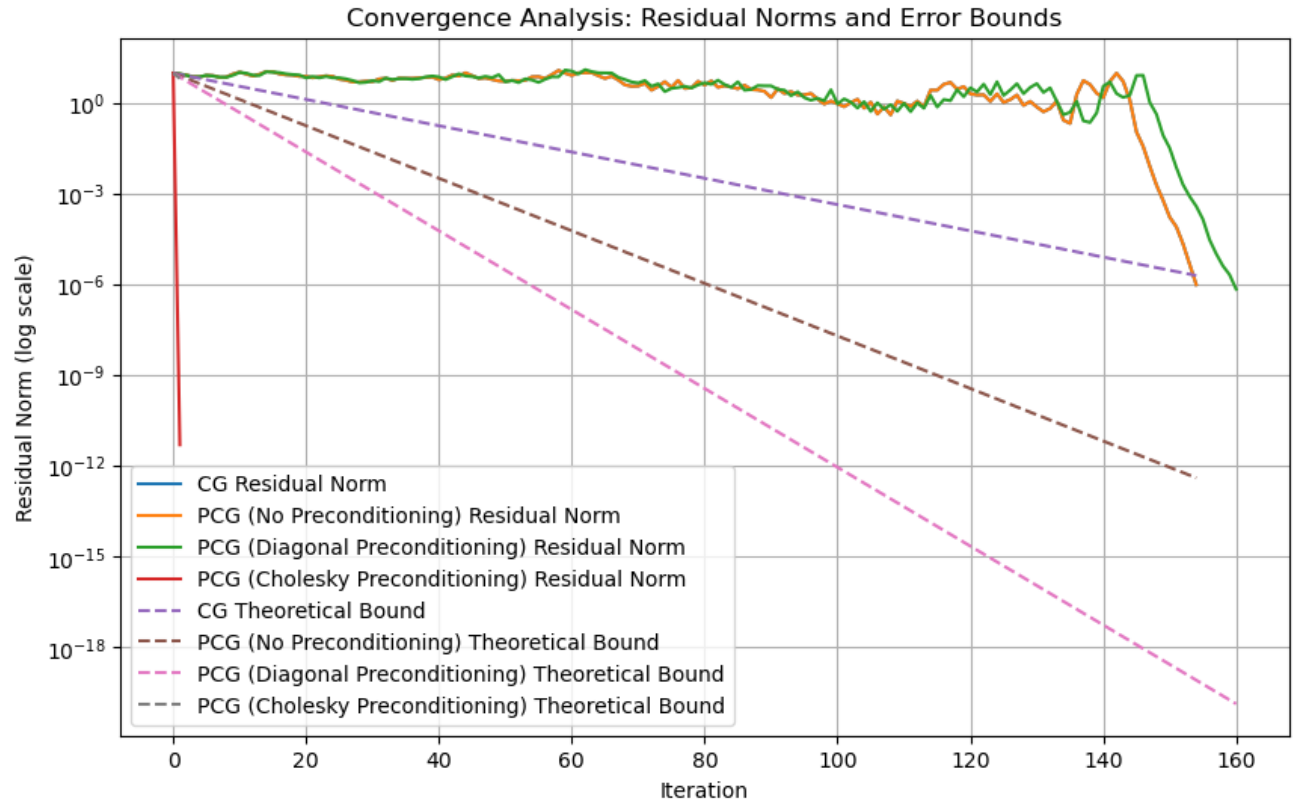
This semi-logarithmic plot confirms the theoretical prediction - notice the approximately linear decrease in log-scale

Augmented Block-CG achieves superior convergence because it effectively improves the condition number of the problem

While theory predicts convergence within n iterations in exact arithmetic, we typically see much faster practical convergence

These residual norms directly translate to solution accuracy - each order of magnitude decrease represents a 10x improvement in precision

The gap between these curves quantifies the computational savings from augmentation - achieving the same accuracy with significantly fewer matrix-vector products



GAUSSIAN SAMPLING VIA LANCZOS SQUARE ROOT

Gaussian Sampling Goal: Given $\mathbf{b} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ we wish to compute:

Several applications in data science and statistics require sampling Gaussians with a given mean and covariance & Jegelka, 2017). A standard approach is to transform an isotropic Gaussian vector. Indeed, suppose $\mathbf{b} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ (i.e. that entries of $\mathbf{b}$ are independent standard Gaussians). Then, for positive definite $\mathbf{A}$,

$$\boldsymbol{\mu} + \mathbf{A}^{1/2} \mathbf{b} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{A}). \quad (24)$$

In other words, in order to sample Gaussian vectors with covariance $\mathbf{A}$, it suffices to apply the matrix-square root of $\mathbf{A}$ to a standard Gaussian vector and then shift this result by $\boldsymbol{\mu}$. The most computationally difficult part of this is applying the square root of $\mathbf{A}$ to $\mathbf{b}$.

Recall the expression (4):

$$\mathbf{A}^{1/2} \mathbf{b} = \frac{2}{\pi} \int_0^\infty \mathbf{A}(\mathbf{A} + z^2 \mathbf{I})^{-1} \mathbf{b} \, dz. \quad (4)$$

When t is sufficiently large, we might hope that $\text{cg}_t(z^2) \approx (\mathbf{A} + z^2 \mathbf{I})^{-1} \mathbf{b}$. This motivates the following definition.

Lanczos Square Root Iteration (Definition 6.1):

Definition 6.1. The t -th Lanczos square root iteration is defined as

$$\text{sq}_t := \frac{2}{\pi} \int_0^\infty \mathbf{A} \text{cg}_t(z^2) \, dz.$$

Definition 6.2. Let $\mathbf{B} = [\mathbf{b}^{(1)} \dots \mathbf{b}^{(m)}]$. The t -th block-Lanczos square root iterate is defined as

$$\text{bsq}_t^{(i)} := \frac{2}{\pi} \int_0^\infty \mathbf{A} \text{bcg}_t^{(i)}(z^2) \, dz.$$

By using block-Lanczos methods, one can efficiently sample multiple Gaussian vectors, significantly reducing the number of matrix-vector products compared to single-vector methods.

Numerical Experiments: Comparison

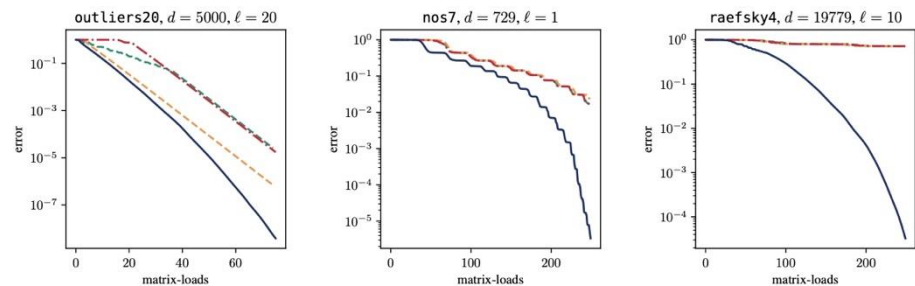
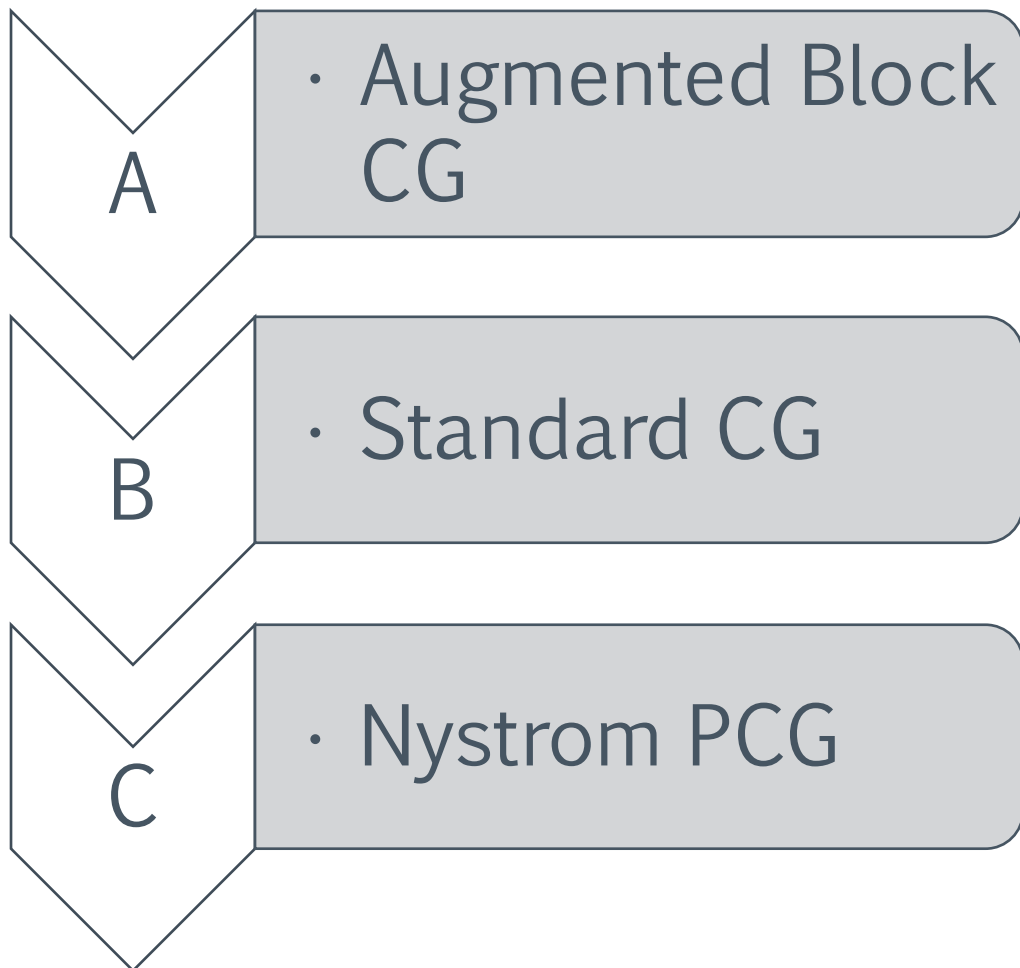


Figure 2. Error versus matrix-loads for block-CG (—), CG (- -), and Nystrom PCG with $s = 1$ (- -) and $s = 3$ (- -) on several test problems.

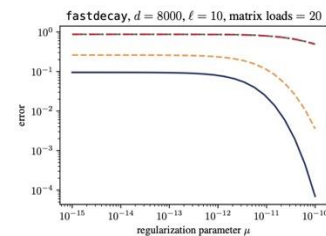


Figure 3. Error after a fixed number of matrix-loads as a function of the regularization parameter μ for block-CG (—), CG (- -), and Nystrom PCG with $s = 1$ (- -) and $s = 3$ (- -).

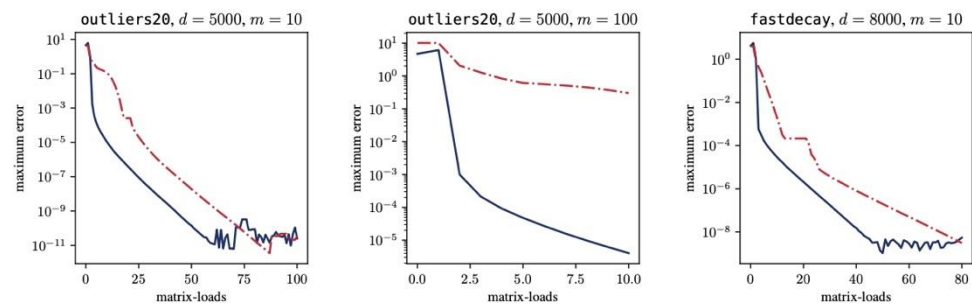


Figure 4. Maximum sample error versus matrix-loads for Lanczos square root (- -) and block-Lanczos square root (—).

COMPARISON

- Augmented block-CG outperforms standard CG and Nystrom PCG.
- The method efficiently computes solutions for multiple μ values (beneficial for the entire ridge-regression path).
- In Gaussian sampling experiments, the block-Lanczos square root method yields significantly lower error for a given number of matrix-vector products.

CONVERGENCE GUARANTEE FOR GAUSSIAN SAMPLING

Convergence Guarantee for
Gaussian Sampling:

› Theorem:

Then, with probability at least $1 - \delta$,

$$\left\{ \forall i : \frac{\|\mathbf{A}^{1/2} \mathbf{b}^{(i)} - \mathbf{bsq}_t^{(i)}\|}{\|\mathbf{A}^{1/2}\| \|\mathbf{b}^{(i)}\|} \leq \varepsilon_t \right\}.$$

This guarantee shows that the block-Lanczos square root iteration converges rapidly, ensuring accurate sampling from $N(\mu, \mathbf{A})$

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